Physics 4140H: Assignment II

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Problem 1. By using the Euler equations with an auxiliary condition imposed, find the shortest path between any two points on the surface of a sphere.

Solution 1. We impose the constraint that $g(x) = x^2 + y^2 + z^2 - R^2 = 0$, i.e. that we are on the surface of a sphere of radius R. Euler's equations are

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) + \lambda(x)\frac{\partial g}{\partial y} = 0$$
$$\frac{\partial f}{\partial z} - \frac{d}{dx}\left(\frac{\partial f}{\partial z'}\right) + \lambda(x)\frac{\partial g}{\partial z} = 0$$

where f takes the usual form for distance-measure, $f = \sqrt{1 + (y')^2 + (z')^2}$. Hence, the first two derivatives are zero and

$$f_{y'} = \frac{y'}{\sqrt{1 + (y')^2 + (z')^2}}$$

and

$$f_{z'} = \frac{z'}{\sqrt{1 + (y')^2 + (z')^2}}$$

and $g_y = 2y$ and $g_z = 2z$. So, Euler's equations become

$$\frac{d}{dx}\left(\frac{y'}{\sqrt{1 + (y')^2 + (z')^2}}\right) = 2y\lambda(x)$$
$$\frac{d}{dx}\left(\frac{z'}{\sqrt{1 + (y')^2 + (z')^2}}\right) = 2z\lambda(x)$$

which we can rewrite as a single equation to arrive at

$$\frac{1}{y}\frac{d}{dx}\left(\frac{y'}{\sqrt{1 + (y')^2 + (z')^2}}\right) = \frac{1}{z}\frac{d}{dx}\left(\frac{z'}{\sqrt{1 + (y')^2 + (z')^2}}\right)$$

this is tantalizingly close to something nice but I can't think of a way to incorporate the $1/y$ and $1/z$ so I could get rid of the derivatives, so we'll take the derivatives in full detail. So,

$$\frac{d}{dx}\left(\frac{y'}{\sqrt{1 + (y')^2 + (z')^2}}\right) = \frac{y''\sqrt{1 + (y')^2 + (z')^2} - \frac{y'(y'y'' + z'z'')}{\sqrt{1 + (y')^2 + (z')^2}}}{1 + (y')^2 + (z')^2}$$

and similarly

$$\frac{d}{dx}\left(\frac{z'}{\sqrt{1 + (y')^2 + (z')^2}}\right) = \frac{z''\sqrt{1 + (y')^2 + (z')^2} - \frac{z'(y'y'' + z'z'')}{\sqrt{1 + (y')^2 + (z')^2}}}{1 + (y')^2 + (z')^2}$$

and so the equality becomes, multiplying by yz on either side and cancelling the common denominator,

$$zy''\sqrt{1 + (y')^2 + (z')^2} - \frac{zy'(y'y'' + z'z'')}{\sqrt{1 + (y')^2 + (z')^2}} = yz''\sqrt{1 + (y')^2 + (z')^2} - \frac{yz'(y'y'' + z'z'')}{\sqrt{1 + (y')^2 + (z')^2}}$$

multiplying by the radical

$$zy'' + \cancel{z(y')^2y''} + z(z')^2y'' - \cancel{z(y')^2y''} - zz'z''y' = z''y + y(y')^2z'' + \cancel{(z')^2z''y} - z'yy'y'' - \cancel{(z')^2z''y}$$

so

$$zy'' + z(z')^2y'' - zz'z''y' = z''y + y(y')^2z'' - z'yy'y''$$

now we want to obtain, using the hint, an expression involving x. The only way we can get that is by using the $g(x)$ condition. Note that

$$\frac{dg}{dx} = x + yy' + zz' = 0 \text{ or } -x = yy' + zz'$$

so let's factor the equality to try to obtain $yy' + zz'$. Rearranging,

$$zy'' + z(z')^2y'' + z'yy'y'' = z''y + y(y')^2z'' + zz'z''y'$$

and factoring out the second derivatives

$$\left(z + z(z')^2 + z'yy'\right)y'' = \left(y + y(y')^2 + zz'y'\right)z''$$

and pulling out the extra z' and y' terms from the second two terms in each set of parentheses,

$$\left(z + \left[zz' + yy'\right]z'\right)y'' = \left(y + \left[yy' + zz'\right]y'\right)z''$$

and so using the derivative of the constraint,

$$\left(z - xz'\right)y'' = \left(y - xy'\right)z''.$$

Yay! Now to show that the equation of a plane passing through the origin, $Ax + By = z$, is a solution here. Note that $z' = A + By'$ and $z'' = By''$. Then $y = \frac{1}{B}(z - Ax)$, $y' = \frac{1}{B}(z' - A)$ and $y'' = \frac{1}{B}z''$ so plugging these in,

$$(Ax + By - x(A + By'))\left(\frac{1}{B}z''\right) = \left(\frac{1}{B}(z - Ax) - x\left(\frac{1}{B}(z' - A)\right)\right)By''.$$

Multiplying things out

$$(\cancel{Ax} + By - \cancel{Ax} - Bxy')\left(\frac{1}{B}z''\right) = \left(\frac{1}{B}z - \frac{\cancel{A}}{\cancel{B}}x - x\frac{1}{B}z' + \frac{\cancel{A}}{\cancel{B}}x\right)By''$$

and clearly by cancelling the remaining B factors

$$\left(y - xy'\right)z'' = \left(z - xz'\right)y'',$$

our original equation, reversed. This is a true statement, so the plane must satisfy the equation. The other way to see this is by noting that, since the the plane is linear in x, the second derivatives of y and z are zero and so, trivially, $0 = 0$.

Problem 2. Find the shortest path between (x, y, z) points $(0, -1, 0)$ and $(0, 1, 0)$ on the conical surface $z = 1 - \sqrt{x^2 + y^2}$. What is the length of the path? (Hint: use cylindrical coordinates).

Solution 2.

Problem 3.

(a) Find the curve $y(x)$ that passes through the endpoints $(0, 0)$ and $(1, 1)$ and minimizes the functional

$$I(y) = \int_0^1 \left[\left(\frac{dy}{dx}\right)^2 - y^2\right]dx$$

(b) What is the minimum value of the integral?

(c) Evaluate $I(y)$ for a straight line $y = x$ between points (0,0) and (1,1).

Solution 3.

(a) We immediately apply the Euler-Lagrange equation with $f = \left(\frac{dy}{dx}\right)^2 - y^2$,

$$\frac{\partial f}{\partial y} = -2y, \quad \frac{\partial f}{\partial y'} = 2y', \quad \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 2y''$$

and so

$$y + y'' = 0.$$

This has general solution $y = C_1\cos(x) + C_2\sin(x)$ to which we apply $y(0) = C_1 = 0$ and $y(1) = C_2\sin(1) = 1 \implies C_2 = 1/\sin(1)$, hence

$$y(x) = \frac{1}{\sin(1)}\sin(x)$$

(b) Evaluating it using the function we found above

$$\begin{aligned} I &= \frac{1}{\sin^2(1)}\int_0^1 \cos^2(x) - \sin^2(x)dx \\ &= \frac{1}{\sin^2(1)}\int_0^1 1 - 2\sin^2(x)dx \\ &= \frac{1}{\sin^2(1)}\int_0^1 \cos(2x)dx \\ &= \frac{\sin(2)}{2\sin^2(1)} \approx 0.64. \end{aligned}$$

(c)

$$\begin{aligned} I &= \int_0^1 1 - x^2 dx \\ &= 2/3 \end{aligned}$$

which, as we would hope, is greater than the minimum value we found in (b).

Problem 4. What curve on the surface $z = x^{3/2}$ joining the points $(0, 0, 0)$ and $(1, 1, 1)$ has the shortest arc length?

Solution 4.